Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
PI Name:	Szofran, Frank		
Project Title:	Reduction of Defects in Germanium-Silicon (RDGS)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Glasses and ceramics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Marshall Space Flight Center		
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City:	Huntsville	**State:**	AL
Zip Code:	35812	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-02
Start Date:	02/01/1998	**End Date:**	01/01/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Szofran, Frank	**Contact Phone:**	256-544-7777
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Benz, K. (University of Freiburg Germany) Cobb, S. (NASA Marshall Space Flight Center) Croell, A. (University of Freiberg Germany) Dold, P. (University of Freiberg Germany) Motakef, S. (CAPE Simulations) Volz, M. (NASA Marshall Space Flight Center) Walker, J. (University of Illinois)		
Grant/Contract No.:	Intramural		
Performance Goal No.:	1.25		
Performance Goal Text:	Use microgravity to establish and improve quantitative and predictive relationships between the structure, processing, and properties of materials.		

Task Description:	To achieve the Task Objective, the following will be done or have been done: * measure the relevant material parameters such as contact angle, growth angle, surface tension, and wetting behavior of the GeSi-melt on potential crucible materials; * determine the mechanism of detached growth including the role of convection; * quantitatively determine the differences in defects and impurities among normal Bridgman, detached Bridgman, and float zone (FZ) growth; * investigate the influence of defined azimuthal or meridional flow due to rotating magnetic fields on the characteristics of detached growth; * control time-dependent Marangoni convection in the case of FZ-growth by the use of a rotating magnetic field to examine the influence on the curvature of the solid-liquid interface and the heat and mass transport; and * grow high quality GeSi single crystals with Si concentrations up to 10 at% and diameters up to 20 mm. Determine the necessary conditions for reproducibly achieving detachment of Ge-Si alloys (up to 10% Si) in the Bridgman growth method, and quantitatively compare the defect structures and impurity levels of detached Bridgman, normal Bridgman, and floating-zone grown materials. It is well established that crystals grown without contact with a container have far superior quality to otherwise similar crystals grown in direct contact with a container. In addition to float-zone processing, detached-Bridgman growth is often cited as a promising tool to improve crystal quality, without the limitations of float zoning. Detached growth has been found to occur quite often during microgravity experiments and considerable improvements of crystal quality have been reported for those cases. However, no thorough understanding of the process or quantitative assessment of the quality improvements existed prior to the investigation. This project will determine the means to reproducibly grow Ge-Si alloys in the detached mode. Ge-Si alloys are of technological interest because with lower defect density materials it would be possible to make thermal neutron and x-ray optics, faster devices than with silicon, higher efficiency solar cells, and higher thermopower devices. Although these alloys are model materials in the sense that these results should be applicable to many other materials, the major technological significance is that the same production equipment that is currently used for silicon device manufacture can also be used for germanium-silicon devices. This means that using germanium-silicon instead of some other material to achieve improved device performance will save an immense amount of money in development and purchase of production equipment. [Ed. note: Project continued with Principal Investigator Dr. Martin Volz; project with Dr. Volz covered March 2008-October 2019, per F. Szofran/MSFC and M. Volz/MSFC]
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	See Bibliography section for publications from this project. Ed. note: Project continued with Principal Investigator Dr. Martin Volz; project with Dr. Volz covered March 2008-October 2019, per F. Szofran/MSFC and M. Volz/MSFC. See that project for subsequent reporting.
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